

PowerShell Alias Reference (Python & Git Focus)

This document provides a categorized and compressed reference of PowerShell aliases, with emphasis on workflows relevant to **Python development** and **Git usage**.

How to Use This Document

Each section below contains a set of aliases relevant to a particular task (e.g., file management, content output, etc.). To reduce clutter:

- Click on any "**Click to view aliases**" line to expand or collapse that section.
- Inside each section is a table listing the alias, full command, and a short description.

Understanding `gal` (Get-Alias)

The `gal` alias is short for:

`gal`  `Get-Alias`

Why It's Important

Use Case	Benefit
Discover aliases	Quickly look up what command an alias points to
Translate scripts	Map shorthand (like <code>ls</code> , <code>rm</code>) to full PowerShell cmdlets
Find custom aliases	Check what user-defined shortcuts are active in your session
Reverse lookup	Use <code>gal -Definition CommandName</code> to find aliases for any cmdlet

Examples

Below are a few essential `gal` use cases that can speed up your scripting, debugging, and exploration in PowerShell:

Industry Production Scenarios

These are helpful for DevOps, automation, and production pipeline scenarios:

```
# Check if a CI/CD script alias exists for triggering a pipeline
Set-Alias run-ci "& .\scripts\trigger-build.ps1"
gal run-ci

# Verify aliases for test automation or deployment wrappers
gal runtests
gal deploytest

# Audit all session aliases and export them for documentation or audits
Get-Alias | Where-Object { $_.Source -eq 'Alias' } | Export-Csv .\alias-audit.csv

# Check for deprecated or legacy aliases in enterprise scripts
gal oldtask
gal -Definition Out-Null

# Open PowerShell profile to inspect or define persistent aliases
notepad $PROFILE
```

General Alias Discovery

Below are a few essential `gal` use cases that can speed up your scripting, debugging, and exploration in PowerShell:

```
# Look up what 'ls' really does
gal ls

# Show all aliases for Get-Content
gal -Definition Get-Content

# View all defined aliases, sorted alphabetically
gal | Sort-Object Name
```

Tip: Use `gal` regularly when learning PowerShell, troubleshooting scripts, inspecting custom environments, or reverse-engineering automation pipelines. It acts like a dictionary for your shell — especially useful when switching between systems or inheriting someone else's script.

Git Context Example

```
# Check if any PowerShell aliases are masking Git commands
gal git

# If you've aliased 'gs' or 'glog' in your PowerShell profile, verify them:
gal gs
```

```
gal glog
```

```
# Confirm 'git' is invoking the correct executable  
Get-Command git
```

You might define PowerShell aliases for common Git shortcuts in your profile, like:

```
Set-Alias gs "git status"  
Set-Alias glog "git log --oneline --graph"
```

Then use `gal` to rediscover or verify them during debugging or customization.

File System & Navigation

Alias	Full Command	Description
<code>ls</code>	<code>Get-ChildItem</code>	List items in a directory
<code>dir</code>	<code>Get-ChildItem</code>	Same as <code>ls</code>
<code>gci</code>	<code>Get-ChildItem</code>	More verbose listing
<code>cd</code>	<code>Set-Location</code>	Change directory
<code>sl</code>	<code>Set-Location</code>	Shortcut to <code>cd</code>
<code>pwd</code>	<code>Get-Location</code>	Show current directory
<code>..</code>	<code>Set-Location ..</code>	Go up one directory
<code>pushd</code>	<code>Push-Location</code>	Save current location
<code>popd</code>	<code>Pop-Location</code>	Return to last saved location

File Operations

Alias	Full Command	Description
<code>cp</code>	<code>Copy-Item</code>	Copy files
<code>copy</code>	<code>Copy-Item</code>	Same as <code>cp</code>
<code>mv</code>	<code>Move-Item</code>	Move or rename files
<code>move</code>	<code>Move-Item</code>	Same as <code>mv</code>

Alias	Full Command	Description
<code>rm</code>	Remove-Item	Delete file or folder
<code>del</code>	Remove-Item	Same as <code>rm</code>
<code>erase</code>	Remove-Item	Same as <code>rm</code>
<code>ni</code>	New-Item	Create new file/folder
<code>md</code>	New-Item	Also used to create folders
<code>mkdir</code>	New-Item	Directory creation
<code>ri</code>	Remove-Item	Remove file/folder
<code>rmdir</code>	Remove-Item	Remove directories

Content & Output

Alias	Full Command	Description
<code>cat</code>	Get-Content	Show file contents
<code>type</code>	Get-Content	Same as <code>cat</code>
<code>gc</code>	Get-Content	More concise
<code>echo</code>	Write-Output	Print to screen
<code>write</code>	Write-Output	Same as <code>echo</code>
<code>tee</code>	Tee-Object	Output to console <i>and</i> file
<code>sc</code>	Set-Content	Overwrite file content
<code>add-content</code>	Add-Content	Append to file
<code>out</code>	Out-Default	Pipe data to output
<code>out-null</code>	Out-Null	Suppress output

Search & Filter

Alias	Full Command	Description
<code>%</code>	ForEach-Object	Loop or map-like ops
<code>foreach</code>	ForEach-Object	Same as <code>%</code>

Alias	Full Command	Description
<code>?</code>	<code>Where-Object</code>	Filter elements
<code>where</code>	<code>Where-Object</code>	Same as <code>?</code>
<code>sls</code>	<code>Select-String</code>	Grep-like search
<code>findstr</code>	<code>Select-String</code>	Alias for compatibility with CMD users

System & Processes

Alias	Full Command	Description
<code>ps</code>	<code>Get-Process</code>	List running processes
<code>gps</code>	<code>Get-Process</code>	Same as <code>ps</code>
<code>kill</code>	<code>Stop-Process</code>	End a process
<code>sp</code>	<code>Set-ItemProperty</code>	Set registry or config value
<code>gp</code>	<code>Get-ItemProperty</code>	Get registry or config value
<code>gsv</code>	<code>Get-Service</code>	List services
<code>start</code>	<code>Start-Process</code>	Open a file or run app
<code>ii</code>	<code>Invoke-Item</code>	Open file in default program

Miscellaneous / Niche

Alias	Full Command	Description
<code>clc</code>	<code>Clear-Content</code>	Clear a file's content
<code>measure</code>	<code>Measure-Object</code>	Count lines/words/etc
<code>select</code>	<code>Select-Object</code>	Choose fields from output
<code>sort</code>	<code>Sort-Object</code>	Sort output
<code>group</code>	<code>Group-Object</code>	Group by property
<code>limit</code>	<code>Select-Object -First</code>	Limit rows from output
<code>history</code>	<code>Get-History</code>	Command history
<code>h</code>	<code>Get-History</code>	Same as above

Alias	Full Command	Description
<code>r</code>	<code>Invoke-History</code>	Re-run previous command
<code>read</code>	<code>Read-Host</code>	Prompt user for input
<code>ipconfig</code>	<code>Get-NetIPConfiguration</code>	Modern networking cmd

View All Current Aliases in PowerShell

```
Get-Alias | Sort-Object Name
```

To export them:

```
Get-Alias | Export-Csv aliases.csv -NoTypeInfo
```